IBM Environmental Intelligence Suite

By IBM Environmental Intelligence Suite

IBM Environmental Intelligence Suite (EIS) is a cloud-based climate and sustainability platform that combines proprietary and third party data, including geospatial data, to derive business insights.

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Getting Started with APIs

Welcome to the IBM Environmental Intelligence Suite (EIS). EIS includes APIs for:

1. Carbon Performance Engine
2. Outage Prediction
3. Geospatial Analytics
4. History on Demand - Direct API
5. Historical Data for Analytical Tools API
6. Forecast Data for Analytical Tools API
7. History on Demand - Utility API
8. History on Demand - Usage API

Geospatial Query Service
History on Demand - Direct API
Historical Data for Analytical Tools
Forecast Data for Analytical Tools
History on Demand - HoD - Utility API
History on Demand - Usage API
Outage Prediction API
FAQ and support

This Getting Started guide will show you how to get started with respective component APIs using an EIS Core API Key by following the steps below:

- ⓘ As an alternative to the EIS Core API Key flow, if you already have an IBM Cloud account membership and wish to use your IBM Cloud Identity and Access Management (IAM) API key follow steps in the IBM Cloud IAM Token Exchange developer guide.

Get Started Steps

1. Get an API Key
2. Provide authentication
3. Specify common API parameters
4. Finally, Use the EIS APIs to make calls to one of the EIS component APIs; such as these below:
   - **Carbon Performance:** Calculates location based emission
     - For Carbon Performance Engine API authentication, see section *Use the EIS APIs - Carbon Performance Engine*.
   - **Outage Prediction:** Forecast outage area
   - **Geospatial Analytics:** Submit query
   - **History on Demand:** Direct API

Get an API Key

1. **Create an IBMid at the IBMid sign up page.**
   - Each team member using EIS component APIs will need their own IBMid.
2. **Share the IBMidemail address with IBM.**
   1. The email address used for creating your IBMid is given to your IBM account representative or salesperson.
   2. If there will be a group of users who require EIS API access, it is suggested each IBMId email is shared together as a set.
3. **IBM will email your EIS Core API Key.**

1. Accounts for each user will be provisioned by IBM and an invitation email is sent which includes an API Key..

4. **Once the invitation is received, users can:**
    1. Login to EIS with their IBMId email at https://www.app.ibm.com/launch/
    2. Use their API Key for the authentication step of EIS API component usage.
5. ***Next:*** authentication

API Authentication

EIS component APIs authenticate requests with a JWT using the HTTP Authorization header Bearer scheme value e.g.

`Authorization: Bearer JWT`

Where:

- `JWT` = JSON Web Token (JWT)

To get a token for API request authentication, you will need your:

1. `orgId`
2. `X-IBM-Client-Id` for `saascore`
3. `X-API-Key` (*Get an API Key*)

Use the Platform Authentication API with the parameters listed above get a token.

Get a token

```
curl --silent --request GET \
  --url 'https://api.ibm.com/saascore/run/authentication-retrieve/api-key?orgId=REPLACE_THIS_VALUE' \
  --header 'X-Api-Key: REPLACE_THIS_KEY' \
  --header 'X-IBM-Client-Id: REPLACE_THIS_KEY'
```

- *See common API parameters for a description of API parameters.*

Next: Use the EIS APIs

Use the EIS APIs

The JWT that you obtain by getting a token is then used as the HTTP Authorization header Bearer scheme value along with the respective API provider name clientId to make EIS API requests.

Geospatial Analytics

Follow link above to see other code examples.

Parameters

- `{geo}`
- `X-IBM-Client-Id`
- `Bearer`

```
curl --request POST \
  --url
https://api.ibm.com/geospatial/run/{geo}/core/v3/query \
  --header 'Authorization: Bearer REPLACE_BEARER_TOKEN' \
  --header 'X-IBM-Client-Id: REPLACE_THIS_KEY' \
  --header 'accept: application/json' \
  --header 'content-type: application/json' \
  --data REPLACE_REQUEST_BODY
```

History On Demand - Direct API

Follow link above to see other code examples.

Parameters

- `{geo}`
- `X-IBM-Client-Id`
- `Bearer`

```
curl --request GET \
  --url 'https://api.ibm.com/geospatial/run/v3/wx/hod/r1/direct
geocode=REPLACE_THIS_VALUE&startDateTime=REPLACE_THIS_VALUE&enc
\
  --header 'Authorization: Bearer REPLACE_BEARER_TOKEN' \
  --header 'X-IBM-Client-Id: REPLACE_THIS_KEY' \
  --header 'accept: application/json'
```

Outage Prediction

Follow link above to see other code examples.

Parameters

- `{geo}`
- `X-IBM-Client-Id`
- `Bearer`

```
curl --request GET \
  --url
'https://api.ibm.com/outageprediction/run/REPLACE_GEO/v3/foreca
format=REPLACE_THIS_VALUE' \
  --header 'Authorization: Bearer REPLACE_BEARER_TOKEN' \
  --header 'X-IBM-Client-Id: REPLACE_THIS_KEY' \
  --header 'accept: application/json'
```

Carbon Performance Engine

Follow link above to see other code examples.

Note: clientId not applicable for Carbon Performance Engine APIs.

Parameters

- `Bearer`
  - A `Bearer` token for the Carbon Performance Engine is obtained in a different manner to that of other EIS component APIs at this time. Please see the Carbon Accounting Developer Guide for details.

```
curl --request POST \
  --url https://foundation.agtech.ibm.com/v2/carbon/location
\
  --header 'Authorization: Bearer REPLACE_BEARER_TOKEN' \
  --header 'accept: application/json' \
  --header 'content-type: application/json' \
  --data REPLACE_REQUEST_BODY
```

Common EIS Component API Parameters

Organization Id

The organization Id is a universally unique identifier (UUID) assigned to you as a customer of IBM software products. An organization Id is submitted as a query parameter for requests to the Platform Authentication provider API (`saascore`).

For example, if your organization Id is `AAAAAAAA-AAAA-AAAA-AAAA-AAAAAAAAAAAA` the authentication request will include the query paramter:

`orgId=AAAAAAAA-AAAA-AAAA-AAAA-AAAAAAAAAAAA`

- Your organization Id can be viewed in the EIS UI: API Information Page.

Client Id

A client Id value is provided with API requests as the HTTP header `X-IBM-Client-Id` and takes the form of:

- `provider-tenantId`

Where:

- `provider` is the EIS API provider name
- `tenantId` is a universally unique identifier (UUID) assigned to you by IBM as an API client of EIS. An Organization may have multiple `tenantId` values.

The client Id value is set to the value of the HTTP header:

- `X-IBM-Client-Id`

For example, if your **tenantId** is `00000000-0000-0000-0000-000000000000`,

Provider API	Name	ClientId	Header
Platform Authentication	`saascore`	`saascore-00000000-0000-0000-0000-000000000000`	`X-IBM-C saascore 0000000( 0000-00( 0000000(`

Provider API	Name	ClientId	Header
Geospatial Analytics	`geospatial`	`geospatial-00000000-0000-0000-0000-000000000000`	`X-IBM-C geospati 0000000( 0000-00( 0000000(`
History on Demand	`geospatial`	`geospatial-00000000-0000-0000-0000-000000000000`	`X-IBM-C geospati 0000000( 0000-00( 0000000(`
Outage Prediction	`outageprediction`	`outageprediction-00000000-0000-0000-0000-000000000000`	`X-IBM-C outagepr 0000000( 0000-00( 0000000(`
Carbon Performance Engine	N/A	N/A	N/A, `X-I Id` not cu required.

- An EIS organization administrator on your team can view provider API client Ids in the EIS UI: API Information Page.

Geographic Region

The geographic region parameter is for future use. This parameter could be used for API request isolation and/or network optimization.

Geographic location is an HTTP request path parameter.

Use the default value of: `na`. For example:

- `https://api.ibm.com/geospatial/run/na`

Open API Specification

About specification

API Version

1.0

Legend

Categories

Analytics

Company information

IBM Environmental Intelligence Suite